Annex IV. External Debt Sustainability

The update of the Joint Fund-Bank Debt Sustainability Analysis for Low-Income Countries maintains Comoros' risk of external debt distress at moderate. Both private and public net current transfers are sizeable in Comoros, and shocks to transfers therefore poses significant risk to debt sustainability. The strong upward external public debt trajectory during the projection period has the potential to become unsustainable without corrective actions. The results also highlight the importance of exchange rate stability and improving external competitiveness for the external debt sustainability outlook, and that any future borrowing must take place exclusively on concessional terms. In these circumstances, simulations suggest that a scenario in which large-scale infrastructure projects are funded by a mix of concessional and non-concessional borrowing would significantly worsen Comoros' debt sustainability outlook. Public sector domestic debt and private sector external debt are minimal in Comoros and are expected to remain so for the foreseeable future.[1]

A. Background

1. Comoros' external public and publicly guaranteed (PPG) debt[2] stock at end 2017 was chiefly held by official bilateral creditors (Text Table 1). Upon reaching the HIPC Completion Point in December 2012, Comoros received extensive irrevocable debt relief, reducing nominal external debt by 22 percentage points of GDP, to 18.5 percent of GDP at end-2013. Since then, it has contracted four external loans, totaling a disbursed amount of US$73.7 million. Two loans were allocated to the construction of a heavy-fuel electricity generation plant and to strengthen domestic telecommunications infrastructure. In addition, two other loans were contracted for rehabilitating the road network: one from the African Development Bank (AfDB) for US$3.7 million and the second from the Saudi Fund for US$30 million, of which only US$0.2 million was disbursed in 2017. The DSA includes the Saudi Fund loan disbursement schedule provided by the authorities for the period 2018-2020. Domestic borrowing by the government is limited to a three-year, KMF 4.8bn (1.6 percent of GDP) loan that was contracted with EXIMBANK in 2017.

Text Table 1: Nominal Stock of External PPG Debt, 2017[1]
(Millions of U.S. Dollars; end-of-period)

Total External Debt[2]	**191.6**
Multilateral Creditors	**76.4**
IMF	14.3
IDA	12.9
BADEA	28.3
Other multilateral creditors	20.9
Official Bilateral	**115.2**
Paris Club Creditors	3.4
Non-Paris Club Creditors	111.8
India	38.9
China	30.9
Kuwait	25.9
Mauritius	0.9

[1] Following Paris Club cancellation of all its HIPC-eligible debt, rescheduling of short-term debt in arrears, and restructuring non-Paris club debt.
[2] Excludes $2.72mn of hospital debt owed by Comoros to France that is the subject of ongoing negotiations.
Source: Comorian authorities.

[1] Comoros' three-year average CPIA score is 2.8, implying a weak policy performance rating for the current DSA.
[2] As in the last DSA exercise, debt is defined as including direct debt and SOEs debt with an explicit state guarantee.

B. Underlying Assumptions

2. Compared to the DSA prepared for the 2016 Article IV consultation, the medium to long-term scenario underlying the present DSA reflects some slight changes in the macroeconomic assumptions, particularly in the growth path (Text Table 2). These changes reflect near-term economic challenges and political difficulties faced by Comoros and ongoing macroeconomic imbalances.

Text Table 2. Comoros: Assumptions for DSA

	2016 Article IV DSA			2018 Article IV DSA		
	2016-2020	2026	2036	2018-2023	2028	2038
GDP Growth Rate (percent)	3.5	4.0	4.0	3.0	3.3	3.3
Inflation, end of period (percent)	2.0	2.0	2.0	1.8	2.0	2.0
Government Revenues and Grants (percent of GDP)	24.3	25.3	24.2	26.4	27.5	28.0
Overall Fiscal Balance (percent of GDP)	-4.7	-6.8	-7.7	-5.6	-8.0	-9.1
FDI (percent of GDP)	1.3	1.6	1.6	1.1	1.6	1.6
Current Account Deficit (Percent of GDP)	10.3	12.0	9.8	7.1	7.7	8.3
Export of Goods and Services (annual percentage growth)	2.8	2.9	3.2	5.3	4.8	4.7
Import of Goods and Services (annual percentage growth)	8.3	3.3	3.8	5.5	5.5	5.7
Net Current transfers (percent of GDP)	19.1	16.9	14.6	20.8	20.9	21.0

- **Real growth** in 2017 was revised downward to 2.7 percent, compared to a projected 3.3 percent in the last DSA. Growth increased between 2016 and 2017 by half a percentage point, but economic activity was slower than expected, even despite the improved electricity supply. This was reflected by the return of isolated power outages and the perceived deterioration of the business climate in parallel with emerging tensions in the financial sector. Also, the baseline medium-term growth rate was slightly revised down from 3.5 percent per year in the last DSA to an average 3 percent, which would eventuate in the absence of continued economic reforms and ongoing political uncertainty associated with the National Conferences.

- Guided by the peg to the euro and the monetary cooperation agreement with France, **inflation** is expected to remain moderate, averaging 2 percent over the long-term.

- **The current account deficit** for 2017 came in at 4.1 percent of GDP, compared to a projected 10.1 percent of GDP in the 2016 DSA. This was largely due to stronger private remittances and grants received from the Middle East (about 3 percent of GDP). Over the medium-to-long term, the current account is projected to increase to about 8 percent on average. This is lower than in the previous DSA and reflects the reduction in growth forecasts and the associated slowdown in consumption, investment, and imports.

- **Net current transfers** allow Comoros to partially balance its large goods and services deficit. Net private remittances increased by 2.4 percentage points to 18.6 percent of GDP in 2017 and are expected to remain stable at around 19 percent of GDP over the medium-to-long term. After recent volatility due to the receipt of budget support, especially from the Middle East, net public transfers are assumed to stabilize at 1.7 percent of GDP over the forecast horizon.

- **Gross investment** increased from 18.4 percent of GDP in 2015 to 21.4 percent in 2017. The forecast assumes a gradual improvement over the medium term to 23.1 percent of GDP in 2023, driven by both public and private investments, and a stabilization at around the same level in the long term.

- **Public investment** was volatile and averaged about 6 percent of GDP over the period 1995-2009. Between 2012 and 2017, public investment averaged 8.9 percent of GDP and is expected to reach 10.7 percent of GDP in 2018. In the medium-term, public investment is expected to increase gradually to 11.3 percent by 2023. Due mainly to the disbursement of the Chinese and Indian loans, net foreign borrowing increased by a total of US$59.6 million (21.8 percent of GDP) between 2015 and 2017. New external borrowing, on concessional terms, is projected to rise from 2.1 percent of GDP in 2018 to 4 percent in 2038. As in the previous DSA exercise, this assumption of increased external borrowing to finance the country's development needs plays a key role in the DSA and explains to some extent the upward slope of the debt burden indicators. This assumption is not based on concrete borrowing plans, but it remains reasonable to expect that Comoros will need to borrow into the future to finance productive infrastructure investments (in roads, tourism, electricity and other sectors). Comoros is also expected to continue to benefit from significant grant financing well into the foreseeable future, leaving the grant element of new borrowing roughly at about 10 percent. No further domestic borrowing by the government is assumed.

- The **overall fiscal deficit,** on a cash basis, is projected to temporarily be limited to 3.1 percent of GDP in 2018 due mainly to the expected receipt of 2 percent of GDP in grants from the World Bank and AFDB. In the baseline scenario; and under the assumption of no further budget support, the fiscal deficit would widen to 6.7 percent of GDP by 2023. The primary balance is expected to remain negative over the horizon, with the deficit mainly financed through external loans for investment purposes.

C. External DSA

3. Private remittances are included in the denominator of the debt and debt service indicators for the purposes of the DSA baseline, due to their importance to the Comorian economy. Between 2009 and 2016, personal remittances averaged 19.4 percent of GDP, ranking Comoros as one of the largest recipients of remittances in Sub-Saharan Africa (SSA). Considering the ongoing political tensions in the aftermath of the National Conferences, the sustainability of remittance inflows constitutes an important source of vulnerability.

4. While there has been marginal increase in the overall debt burden, this DSA update maintains the previous moderate external debt distress rating. As in the previous DSA, all baseline debt and debt service indicators remain below their respective thresholds over the forecast horizon (Figure A1; Tables A1 and A2). However, both debt service indicators exhibit an upward trend from 2025 onwards, reflecting increased debt service commitments and a persistent deterioration in the revenue outlook.

5. The results of the stress tests and alternative scenarios exhibit some similarities and differences with those of the previous DSA. As in the previous DSA, the ratio of debt to GDP plus remittances, as well as the PV of debt to revenue ratio, breach the threshold in the later part of the

projection period under a one-time depreciation shock (in the former case the breach is a sustained one).[3] The susceptibility to a one-time depreciation shock underscores the importance of exchange rate stability brought by the currency peg. A shock to exports, which is now the most extreme shock[4] to the PV of debt to exports plus remittances, results in this ratio breaching the relevant threshold. A shock to the terms of new borrowing, while not the most extreme shock in this case, would result in a sustained threshold breach by the end of the forecast horizon.[5] These results highlight that improving external competitiveness through removing structural impediments to export activity is critically important. Also, vulnerability to a future deterioration in the terms of new borrowing underlines the importance of maintaining a prudent debt management strategy and borrowing exclusively on concessional terms. Under the historical scenario, as in the previous DSA there are no breaches of the different thresholds. The initial improvement in the debt and debt service indicators in the baseline continues to reflect that the current account deficits in the baseline are larger than historical deficits over the medium-term.

6. The policy adjustment scenario (see section: Outlook, Risks and Debt Sustainability in the Staff Report) would see marginal improvements in Comoros' external debt sustainability, although key vulnerabilities related to remittances would remain. Here, growth is projected to accelerate to 4.7 percent in 2023 and to stabilize around this level in the long term. While the current account deficit would average 5.7 percent of GDP over the medium term, the overall fiscal deficit is projected to decrease on average from 5.6 percent of GDP in the baseline to an average 2.3 percent of GDP in the adjustment scenario. Yet vulnerabilities related to the high economic importance of remittances, as well as shocks to exports and a one-time currency depreciation would remain (Figure A3). Under a one-time depreciation shock, the most extreme shocks induce a breach of the threshold of the PV of debt-to-GDP plus remittances. Also, even if not the most extreme shock this threshold is breached under a shock of the terms of new borrowing. Borrowing on non-concessional terms would also continue to pose problems for debt sustainability, but the authorities would have greater room for maneuver were additional financing are on concessional terms.

7. Discussions are underway on financing a project for a large new deep-water port for *Grande Comore* island (US$115 million / 18 percent of 2017 GDP), while plans to finance the rebuilding of El Maarouf hospital (US$50 million/ 8 percent of 2017 GDP) through non-concessional borrowing have reportedly been shelved. A simulation illustrates the implications for debt sustainability of borrowing for such large infrastructure projects (Figure A1 and A3, dotted blue line) in both baseline and policy scenarios, using the latest available project information and conservative assumptions on financing terms that are typical of such projects. For the total amount of US$165 million needed for both projects, a combination of concessional terms for the new deep-water port project and non-concessional terms for a project similar in scale to the hospital are

[3] Defined as a one-time 30 percent nominal depreciation relative to the baseline in 2019.

[4] The most extreme stress test is defined as the test that yields the highest level of debt on or before the tenth year of the projection period.

[5] A shock to exports refers to export value growth at its historical average minus one standard deviation in 2019 – 2020. A shock to new borrowing terms refers to public sector loans on less favorable terms during the period 2018–2038. Less favorable terms are defined as 2 percent increase in the interest rate for new borrowing, while grace and maturity periods are the same as in the baseline.

assumed.[6] The results of this simulation show a significant deterioration in the debt profile in both baseline and policy scenarios, with near-breaches of the thresholds in 2019 for the debt-to-revenue ratio as well as the debt-to-GDP plus remittances. This simulation highlights that large-scale new borrowing could have adverse implications for Comoros' debt sustainability, even assuming a sizable proportion of concessional borrowing.

Authorities' views

8. The authorities acknowledge the importance of maintaining debt sustainability and understand the importance of negotiating external debt on concessional terms, to the extent that such financing is available, and were open to maintaining a purely concessional borrowing policy. Concerning external debt arrears, the authorities maintained that according to their external arrears repayment plan, all external arrears will be paid by the end-March 2019. They further noted that the first repayment installment had already been made.

D. Conclusion

9. The present DSA update maintains Comoros' external debt distress rating at moderate. The debt increases related to the new disbursements have been offset by higher remittances and grants. The results continue to indicate that there are no breaches of debt and debt-service thresholds in the baseline. However, the shocks considered by the DSA framework continue to highlight the economy's high degree of vulnerability to exchange rate instability, and a deterioration in export performance or new borrowing terms. These DSA results underscore the need to strengthen both the business climate to improve both external competitiveness and debt management capacity. It also highlights that future external borrowing, particularly for large infrastructure projects, must be exclusively on concessional terms. The simulation highlights that large-scale new borrowing, even with a sizable proportion on concessional terms, would significantly worsen Comoros' debt sustainability outlook.

[6] This simulation assumes a total concessional loan amount of USD 115 million, with a maturity of 30 years, a grace period of 5 years, and an interest rate of 2.1 percent. Apart from the assumed interest rate, these terms are similar to those of the previously-contracted loan from Exim Bank China and imply a grant element of 35.1 percent. Based on the available information, the simulation assumes for the hospital project an interest rate of 12 percent and a maturity of 12 years.

Table 1. Comoros: External Debt Sustainability Framework, Baseline Scenario, 2015-2018 1/

(in person of GDP, unless otherwise indicated)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2015	2016	2017			2018	2019	2020	2021	2022	2023	2018-2023 Average	2028	2038	2024-2038 Average
xternal debt (nominal) 1/	**22.8**	**26.4**	**28.8**			**27.0**	**27.3**	**27.5**	**27.8**	**28.2**	**28.8**		**32.0**	**38.5**	
of which: public and publicly guaranteed (PPG)	22.8	26.4	28.8			27.0	27.3	27.5	27.8	28.2	28.8		32.0	38.5	
hange in external debt	3.6	3.6	2.3			-1.8	0.3	0.2	0.2	0.4	0.6		0.6	0.5	
lentified net debt-creating flows	2.2	6.0	2.2			4.2	5.9	5.6	5.4	5.1	5.0		5.1	5.5	
Non-interest current account deficit	**-0.3**	**7.3**	**4.1**	**5.2**	**3.2**	**5.7**	**7.5**	**7.2**	**7.1**	**6.8**	**6.7**		**7.3**	**7.7**	7.4
Deficit in balance of goods and services	28.9	25.9	27.8			29.0	28.8	28.6	28.5	28.3	28.4		28.7	28.9	
Exports	16.5	17.9	19.1			18.1	18.2	18.3	18.5	18.8	18.9		17.8	16.0	
Imports	45.4	43.8	46.9			47.0	47.0	47.0	47.0	47.1	47.3		46.5	44.9	
Net current transfers (negative = inflow)	-28.3	-17.6	-23.0	-26.8	5.0	-22.2	-20.3	-20.3	-20.5	-20.6	-20.8		-20.9	-21.0	-20.9
of which: official	-11.1	-1.5	-4.4			-3.6	-1.6	-1.6	-1.6	-1.5	-1.5		-1.7	-1.7	
Other current account flows (negative = net inflow)	-1.0	-1.0	-0.7			-1.0	-1.0	-1.0	-0.9	-0.9	-0.9		-0.6	-0.2	
Net FDI (negative = inflow)	**-0.8**	**-0.4**	**-0.4**	**-1.4**	**1.1**	**-1.1**	**-1.1**	**-1.1**	**-1.1**	**-1.1**	**-1.1**		**-1.6**	**-1.6**	-1.6
Endogenous debt dynamics 2/	**3.4**	**-0.8**	**-1.5**			**-0.4**	**-0.5**	**-0.5**	**-0.6**	**-0.6**	**-0.6**		**-0.6**	**-0.6**	
Contribution from nominal interest rate	0.1	0.1	0.0			0.3	0.2	0.2	0.2	0.2	0.3		0.4	0.6	
Contribution from real GDP growth	-0.2	-0.5	-0.7			-0.7	-0.7	-0.7	-0.8	-0.8	-0.9		-1.0	-1.2	
Contribution from price and exchange rate changes	3.4	-0.5	-0.8			...	...	...	...	...	...		...	...	
esidual (3-4) 3/	**1.4**	**-2.4**	**0.1**			**-6.0**	**-5.5**	**-5.4**	**-5.1**	**-4.7**	**-4.4**		**-4.4**	**-5.0**	
of which: exceptional financing	...	...	...			...	...	...	...	...	...		...	...	
V of external debt 4/	...	...	17.1			15.4	15.7	16.1	16.7	17.4	18.2		22.3	28.4	
In percent of exports	...	...	89.4			85.4	86.3	87.8	90.3	92.8	96.6		125.4	177.3	
V of PPG external debt	**...**	**...**	**17.1**			**15.4**	**15.7**	**16.1**	**16.7**	**17.4**	**18.2**		**22.3**	**28.4**	
In percent of exports	**...**	**...**	**89.4**			**85.4**	**86.3**	**87.8**	**90.3**	**92.8**	**96.6**		**125.4**	**177.3**	
In percent of government revenues	**...**	**...**	**101.5**			**87.4**	**92.8**	**93.9**	**95.8**	**98.4**	**101.3**		**117.2**	**144.2**	
lebt service-to-exports ratio (in percent)	**2.1**	**3.0**	**3.0**			**8.4**	**5.1**	**4.7**	**4.1**	**3.8**	**3.6**		**5.9**	**12.2**	
PG debt service-to-exports ratio (in percent)	**2.1**	**3.0**	**3.0**			**8.4**	**5.1**	**4.7**	**4.1**	**3.8**	**3.6**		**5.9**	**12.2**	
PG debt service-to-revenue ratio (in percent)	**2.0**	**3.7**	**3.4**			**8.6**	**5.5**	**5.0**	**4.4**	**4.0**	**3.7**		**5.5**	**9.9**	
otal gross financing need (Millions of U.S. dollars)	-4.9	45.3	27.6			46.0	58.5	59.6	60.9	61.7	64.0		91.7	197.2	
lon-interest current account deficit that stabilizes debt ratio	-3.9	3.7	1.8			7.5	7.1	7.0	6.8	6.4	6.1		6.6	7.2	
ey macroeconomic assumptions															
eal GDP growth (in percent)	1.0	2.2	2.7	2.1	0.8	2.8	2.8	2.9	3.1	3.2	3.3	3.0	3.3	3.3	3.3
DP deflator in US dollar terms (change in percent)	-15.2	2.2	3.3	1.5	8.0	12.5	3.4	3.5	3.0	3.0	2.7	4.7	2.6	2.6	2.6
ffective interest rate (percent) 5/	0.6	0.5	0.2	0.5	0.2	1.1	0.7	0.8	0.9	0.9	1.0	0.9	1.4	1.7	1.5
rowth of exports of G&S (US dollar terms, in percent)	-22.5	13.1	13.4	6.7	14.7	9.6	6.8	7.4	7.0	7.9	6.9	7.6	4.8	4.7	4.8
rowth of imports of G&S (US dollar terms, in percent)	-20.3	0.5	13.8	5.6	14.4	15.9	6.1	6.5	6.4	6.4	6.6	8.0	5.5	5.7	5.6
rant element of new public sector borrowing (in percent)	...	...	...	...	...	49.8	49.1	42.0	36.1	35.1	35.1	41.2	35.1	35.1	35.1
overnment revenues (excluding grants, in percent of GDP)	16.5	14.5	16.8			17.7	16.9	17.2	17.4	17.7	18.0		19.0	19.7	19.2
id flows (in Millions of US dollars) 7/	88.9	54.6	76.2			101.6	90.0	94.2	99.1	106.4	113.9		158.1	302.5	
of which: Grants	88.9	54.6	76.2			79.8	68.4	72.9	77.4	82.3	87.3		116.0	204.2	
of which: Concessional loans	0.0	0.0	0.0			21.8	21.6	21.3	21.7	24.1	26.6		42.0	98.3	
rant-equivalent financing (in percent of GDP) 8/	...	...	...			12.0	9.9	9.6	9.4	9.4	9.5		9.6	9.8	9.6
rant-equivalent financing (in percent of external financing) 8/	...	...	...			89.2	87.8	86.9	86.0	85.3	84.9		82.8	78.9	81.6
lemorandum items:															
lominal GDP (Millions of US dollars)	588.5	614.2	651.6			753.0	800.3	852.4	905.5	962.7	1021.6		1366.2	2443.2	
lominal dollar GDP growth	-14.3	4.4	6.1			15.6	6.3	6.5	6.2	6.3	6.1	7.8	6.0	6.0	6.0
V of PPG external debt (in Millions of US dollars)			116.5			116.9	126.4	137.8	151.6	168.0	186.7		304.8	694.3	
ᵖVt-PVt-1)/GDPt-1 (in percent)						0.1	1.3	1.4	1.6	1.8	1.9	1.4	2.1	2.1	2.1
ross workers' remittances (Millions of US dollars)	167.8	156.0	169.4			195.3	207.4	220.3	234.7	249.1	264.4		337.5	562.9	
V of PPG external debt (in percent of GDP + remittances)	...	...	13.5			12.3	12.5	12.8	13.2	13.8	14.5		17.9	23.1	
V of PPG external debt (in percent of exports + remittances)	...	...	37.8			35.1	35.6	36.5	37.6	39.0	40.7		52.4	72.7	
lebt service of PPG external debt (in percent of exports + remittances)	...	...	1.3			3.4	2.1	1.9	1.7	1.6	1.5		2.5	5.0	

ources: Country authorities; and staff estimates and projections.

/ Includes both public and private sector external debt.

/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

/ Assumes that PV of private sector debt is equivalent to its face value.

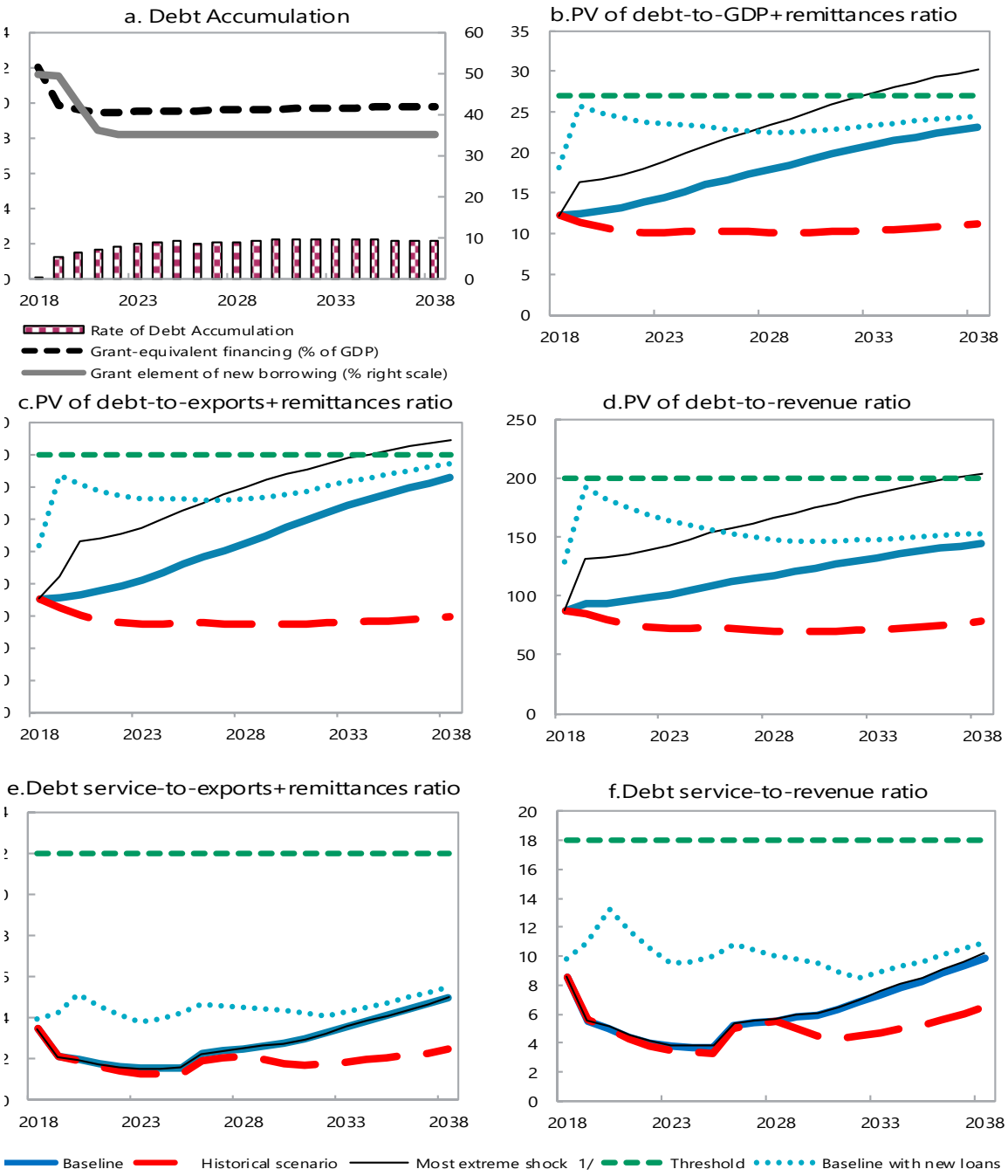
/ Current-year interest payments divided by previous period debt stock.

/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

/ Defined as grants, concessional loans, and debt relief.

/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Figure 1. Comoros: Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios, 2018-2038 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2028. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Growth shock and in figure f. to a Growth shock

Table 2. Comoros: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External debt, 2018-2038 (in percent)

	Projections							
	2018	2019	2020	2021	2022	2023	**2028**	2038
PV of debt-to-GDP+remittances ratio								
Baseline	12	12	13	13	14	14	**18**	23
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2018-2038 1/	12	11	11	10	10	10	**10**	11
A2. New public sector loans on less favorable terms in 2018-2038 2	12	13	14	15	16	17	**22**	33
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2019-2020	12	13	13	14	14	15	**18**	24
B2. Export value growth at historical average minus one standard deviation in 2019-2020 3/	12	14	17	17	17	18	**21**	24
B3. US dollar GDP deflator at historical average minus one standard deviation in 2019-2020	12	14	15	16	16	17	**21**	27
B4. Net non-debt creating flows at historical average minus one standard deviation in 2019-2020 4/	12	12	12	12	13	14	**17**	23
B5. Combination of B1-B4 using one-half standard deviation shocks	12	12	14	15	16	16	**20**	25
B6. One-time 30 percent nominal depreciation relative to the baseline in 2019 5/	12	16	17	17	18	19	**23**	30
PV of debt-to-exports+remittances ratio								
Baseline	35	36	36	38	39	41	**52**	73
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2018-2038 1/	35	32	30	28	28	27	**27**	29
A2. New public sector loans on less favorable terms in 2018-2038 2	35	37	39	42	44	47	**66**	103
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2019-2020	35	36	37	38	39	41	**53**	73
B2. Export value growth at historical average minus one standard deviation in 2019-2020 3/	35	42	53	54	55	57	**70**	84
B3. US dollar GDP deflator at historical average minus one standard deviation in 2019-2020	35	36	37	38	39	41	**53**	73
B4. Net non-debt creating flows at historical average minus one standard deviation in 2019-2020 4/	35	33	33	35	37	39	**50**	72
B5. Combination of B1-B4 using one-half standard deviation shocks	35	32	36	38	40	41	**53**	71
B6. One-time 30 percent nominal depreciation relative to the baseline in 2019 5/	35	36	37	38	39	41	**53**	73
PV of debt-to-revenue ratio								
Baseline	87	93	94	96	98	101	**117**	144
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2018-2038 1/	87	85	80	76	74	73	**70**	78
A2. New public sector loans on less favorable terms in 2018-2038 2	87	97	102	106	112	117	**147**	205
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2019-2020	87	95	97	99	102	105	**121**	149
B2. Export value growth at historical average minus one standard deviation in 2019-2020 3/	87	103	122	123	124	126	**139**	149
B3. US dollar GDP deflator at historical average minus one standard deviation in 2019-2020	87	103	115	117	120	124	**144**	177
B4. Net non-debt creating flows at historical average minus one standard deviation in 2019-2020 4/	87	90	88	90	93	96	**113**	143
B5. Combination of B1-B4 using one-half standard deviation shocks	87	96	113	115	118	121	**139**	167
B6. One-time 30 percent nominal depreciation relative to the baseline in 2019 5/	87	131	133	135	139	143	**165**	204

Table 2. Comoros: Sensitivity Analysis is for Key Indicators of Public and Publicly Guaranteed External Debt, 2018-2038 (Concluded) (in percent)

Debt service-to-exports+remittances ratio								
Baseline	3	2	2	2	2	2	**2**	5.0
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2018-2038 1/	3	2	2	2	1	1	**2**	2
A2. New public sector loans on less favorable terms in 2018-2038 2	3	2	2	2	2	2	**3**	8
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2019-2020	3	2	2	2	2	2	**2**	5
B2. Export value growth at historical average minus one standard deviation in 2019-2020 3/	3	2	2	2	2	2	**3**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2019-2020	3	2	2	2	2	2	**2**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2019-2020 4/	3	2	2	2	2	1	**2**	5
B5. Combination of B1-B4 using one-half standard deviation shocks	3	2	2	2	2	2	**2**	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2019 5/	3	2	2	2	2	2	**2**	5
Debt service-to-revenue ratio								
Baseline	9	5	5	4	4	4	**6**	10
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2018-2038 1/	9	6	5	4	4	3	**5**	6
A2. New public sector loans on less favorable terms in 2018-2038 2	9	5	5	5	5	5	**7**	15
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2019-2020	9	6	5	5	4	4	**6**	10
B2. Export value growth at historical average minus one standard deviation in 2019-2020 3/	9	5	5	5	5	4	**6**	11
B3. US dollar GDP deflator at historical average minus one standard deviation in 2019-2020	9	6	6	5	5	5	**7**	12
B4. Net non-debt creating flows at historical average minus one standard deviation in 2019-2020 4/	9	5	5	4	4	4	**5**	10
B5. Combination of B1-B4 using one-half standard deviation shocks	9	6	6	5	5	4	**6**	12
B6. One-time 30 percent nominal depreciation relative to the baseline in 2019 5/	9	8	7	6	6	5	**8**	14
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	36	36	36	36	36	36	**36**	36

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3. Comoros: Public Sector Debt Sustainability Framework, Baseline Scenario, 2015-2038 (in percent of GDP, unless otherwise indicated)

	Actual					Estimate						Projections			
	2015	2016	2017	Average 5/	Standard Deviation 5/	2018	2019	2020	2021	2022	2023	2018-23 Average	2028	2038	2024-38 Average
Public sector debt 1/	22.8	27.5	29.3			27.1	27.3	27.5	27.8	28.2	28.8		32.0	38.5	
of which: foreign-currency denominated	22.8	26.4	28.8			27.0	27.3	27.5	27.8	28.2	28.8		32.0	38.5	
Change in public sector debt	3.6	4.7	1.8			-2.1	0.2	0.2	0.2	0.4	0.6		0.6	0.5	
dentified debt-creating flows	-2.6	4.2	-4.0			-0.9	3.7	4.2	4.4	4.7	5.1		6.2	6.9	
Primary deficit	-5.9	5.0	-1.2	-3.2	6.5	1.6	5.1	5.6	5.8	6.1	6.4	5.1	7.6	8.5	7.8
Revenue and grants	31.6	23.4	28.5			28.3	25.5	25.7	26.0	26.3	26.6		27.5	28.0	
of which: grants	15.1	8.9	11.7			10.6	8.6	8.6	8.5	8.5	8.6		8.5	8.4	
Primary (noninterest) expenditure	25.7	28.4	27.3			29.9	30.6	31.3	31.8	32.3	33.0		35.1	36.5	
Automatic debt dynamics	3.3	-0.9	-2.8			-2.6	-1.4	-1.4	-1.4	-1.4	-1.3		-1.4	-1.5	
Contribution from interest rate/growth differential	-0.4	-0.9	-1.0			-1.2	-1.1	-1.2	-1.2	-1.2	-1.2		-1.4	-1.5	
of which: contribution from average real interest rate	-0.2	-0.4	-0.3			-0.4	-0.4	-0.4	-0.4	-0.4	-0.3		-0.4	-0.3	
of which: contribution from real GDP growth	-0.2	-0.5	-0.7			-0.8	-0.7	-0.8	-0.8	-0.9	-0.9		-1.0	-1.2	
Contribution from real exchange rate depreciation	3.7	0.0	-1.7			-1.4	-0.3	-0.2	-0.2	-0.1	-0.1		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	6.2	0.5	5.7			-1.2	-3.5	-4.0	-4.2	-4.3	-4.5		-5.5	-6.4	
Other Sustainability Indicators															
PV of public sector debt	...	...	17.6			15.6	15.7	16.1	16.7	17.4	18.2		22.3	28.4	
of which: foreign-currency denominated	...	...	17.1			15.4	15.7	16.1	16.7	17.4	18.2		22.3	28.4	
of which: external	...	...	17.1			15.4	15.7	16.1	16.7	17.4	18.2		22.3	28.4	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	-5.6	5.6	-0.6			3.2	6.0	6.5	6.6	6.8	7.1		8.6	10.4	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	61.7			55.2	61.6	62.7	64.2	66.3	68.7		81.0	101.2	
PV of public sector debt-to-revenue ratio (in percent)	...	...	104.6			88.3	92.8	93.9	95.8	98.4	101.3		117.2	144.2	
of which: external 3/	...	...	101.5			87.4	92.8	93.9	95.8	98.4	101.3		117.2	144.2	
Debt service-to-revenue and grants ratio (in percent) 4/	1.1	2.3	2.0			5.3	3.6	3.3	2.9	2.7	2.5		3.8	6.9	
Debt service-to-revenue ratio (in percent) 4/	2.0	3.7	3.4			8.6	5.5	5.0	4.4	4.0	3.7		5.5	9.9	
Primary deficit that stabilizes the debt-to-GDP ratio	-9.5	0.4	-3.0			3.8	4.9	5.4	5.6	5.6	5.9		6.9	7.9	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	1.0	2.2	2.7	2.1	0.8	2.8	2.8	2.9	3.1	3.2	3.3	3.0	3.3	3.3	3.3
Average nominal interest rate on forex debt (in percent)	0.6	0.5	0.2	0.5	0.2	1.1	0.7	0.8	0.9	0.9	1.0	0.9	1.4	1.7	1.5
Average real interest rate on domestic debt (in percent)	...	...	...			...	...	...	...	...	...		...	...	
Real exchange rate depreciation (in percent, + indicates depreciation)	19.4	0.1	-6.8	1.8	8.1	-5.1	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	1.6	2.4	1.2	3.2	1.4	2.4	2.2	2.3	2.3	2.3	2.3	2.3	2.6	2.6	2.6
Growth of real primary spending (deflated by GDP deflator, in percent	7.0	12.8	-1.4	1.9	4.5	12.7	5.0	5.5	4.5	5.0	5.4	6.4	3.7	3.8	4.0
Grant element of new external borrowing (in percent)	...	...	...	...	...	49.8	49.1	42.0	36.1	35.1	35.1	41.2	35.1	35.1	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

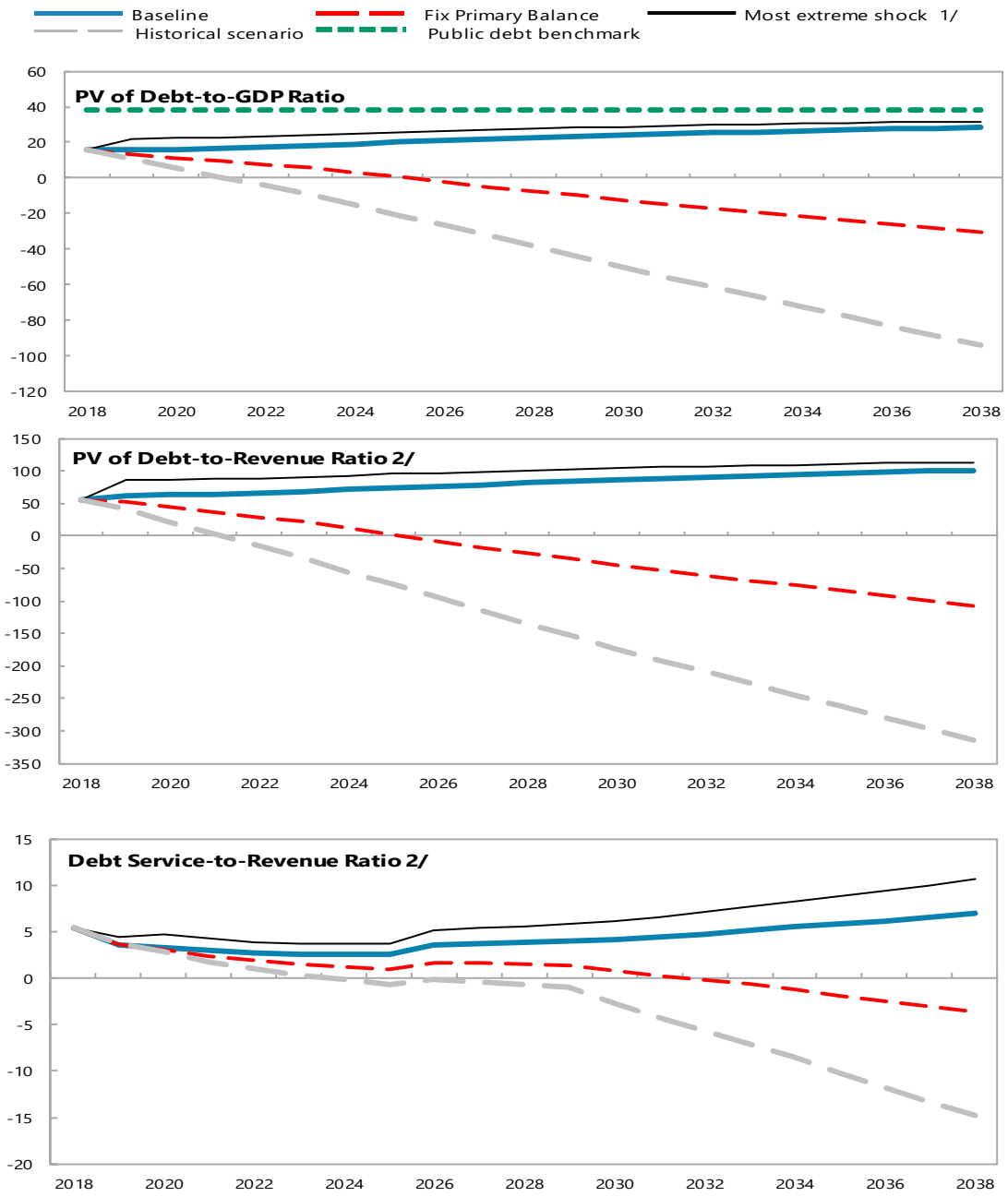
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Comoros: Sensitivity Analysis for Key Indicators of Public Debt 2018-2038
(baseline scenario)

	Projections							
	2018	2019	2020	2021	2022	2023	2028	2038
PV of Debt-to-GDP Ratio								
Baseline	16	16	16	17	17	18	22	28
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	16	11	5	1	-4	-9	-38	-94
A2. Primary balance is unchanged from 2018	16	14	11	9	8	6	-7	-31
A3. Permanently lower GDP growth 1/	16	16	16	17	18	19	24	33
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2019-20	16	16	17	18	19	20	26	35
B2. Primary balance is at historical average minus one standard deviations in 2019-202	16	15	13	14	15	16	20	27
B3. Combination of B1-B2 using one half standard deviation shocks	16	13	10	11	12	13	19	29
B4. One-time 30 percent real depreciation in 2019	16	22	21	21	22	22	25	29
B5. 10 percent of GDP increase in other debt-creating flows in 2019	16	22	22	23	23	24	28	32
PV of Debt-to-Revenue Ratio 2/								
Baseline	55	62	63	64	66	69	81	101
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	55	41	21	2	-16	-35	-135	-314
A2. Primary balance is unchanged from 2018	55	53	44	37	29	21	-27	-109
A3. Permanently lower GDP growth 1/	55	62	63	65	67	70	86	117
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2019-20	55	63	66	69	72	75	93	123
B2. Primary balance is at historical average minus one standard deviations in 2019-202	55	57	52	54	57	59	73	96
B3. Combination of B1-B2 using one half standard deviation shocks	55	49	37	41	44	48	68	103
B4. One-time 30 percent real depreciation in 2019	55	85	83	83	83	84	90	102
B5. 10 percent of GDP increase in other debt-creating flows in 2019	55	87	87	88	89	91	100	113
Debt Service-to-Revenue Ratio 2/								
Baseline	5	4	3	3	3	3	4	7
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	5	4	3	2	1	0	-1	-15
A2. Primary balance is unchanged from 2018	5	4	3	2	2	2	1	-4
A3. Permanently lower GDP growth 1/	5	4	3	3	3	3	4	8
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2019-20	5	4	3	3	3	3	4	8
B2. Primary balance is at historical average minus one standard deviations in 2019-202	5	4	3	3	2	2	4	6
B3. Combination of B1-B2 using one half standard deviation shocks	5	4	3	2	2	2	4	6
B4. One-time 30 percent real depreciation in 2019	5	4	5	4	4	4	6	11
B5. 10 percent of GDP increase in other debt-creating flows in 2019	5	4	4	4	3	3	4	8

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Figure 2. Comoros: Indicators of Public Debt Under Alternative Scenarios, 2018-2038 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2028.
2/ Revenues are defined inclusive of grants.

Table 5. Comoros: External Debt Sustainability Framework, Policy Adjustments Scenario, 2015-2038 1/ (in percent of GDP, unless otherwise indicated)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2015	2016	2017			2018	2019	2020	2021	2022	2023	2018-2023 Average	2028	2038	2024-2038 Average
External debt (nominal) 1/	**22.8**	**26.4**	**28.8**			**26.7**	**26.8**	**26.8**	**27.0**	**27.2**	**27.5**		**29.8**	**35.4**	
of which: public and publicly guaranteed (PPG)	22.8	26.4	28.8			26.7	26.8	26.8	27.0	27.2	27.5		29.8	35.4	
Change in external debt	3.6	3.6	2.3			-2.0	0.1	0.0	0.2	0.2	0.3		0.5	0.5	
Identified net debt-creating flows	2.2	6.0	2.2			4.1	4.0	3.7	3.6	3.3	2.9		3.6	2.6	
Non-interest current account deficit	**-0.3**	**7.3**	**4.1**	**5.2**	**3.2**	**5.7**	**5.9**	**5.6**	**5.5**	**5.3**	**4.9**		**6.1**	**5.2**	5.9
Deficit in balance of goods and services	28.9	25.9	27.8			29.1	29.2	28.9	28.7	28.3	28.0		27.3	25.9	
Exports	16.5	17.9	19.1			18.1	18.2	18.5	18.8	18.9	19.1		18.5	17.8	
Imports	45.4	43.8	46.9			47.2	47.4	47.4	47.5	47.2	47.1		45.8	43.7	
Net current transfers (negative = inflow)	-28.3	-17.6	-23.0	-26.8	5.0	-22.4	-22.3	-22.3	-22.3	-22.2	-22.2		-20.8	-20.6	-20.7
of which: official	-11.1	-1.5	-4.4			-3.6	-3.4	-3.3	-3.2	-3.1	-3.0		-1.7	-1.7	
Other current account flows (negative = net inflow)	-1.0	-1.0	-0.7			-1.0	-1.0	-1.0	-0.9	-0.8	-0.8		-0.5	-0.1	
Net FDI (negative = inflow)	**-0.8**	**-0.4**	**-0.4**	**-1.4**	**1.1**	**-1.1**	**-1.1**	**-1.1**	**-1.1**	**-1.1**	**-1.1**		**-1.6**	**-1.6**	-1.6
Endogenous debt dynamics 2/	**3.4**	**-0.8**	**-1.5**			**-0.6**	**-0.8**	**-0.8**	**-0.8**	**-0.9**	**-0.9**		**-0.9**	**-1.0**	
Contribution from nominal interest rate	0.1	0.1	0.0			0.3	0.2	0.2	0.2	0.2	0.3		0.4	0.6	
Contribution from real GDP growth	-0.2	-0.5	-0.7			-0.8	-1.0	-1.0	-1.0	-1.1	-1.2		-1.3	-1.5	
Contribution from price and exchange rate changes	3.4	-0.5	-0.8			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**1.4**	**-2.4**	**0.1**			**-6.1**	**-3.9**	**-3.7**	**-3.4**	**-3.2**	**-2.6**		**-3.1**	**-2.1**	
of which: exceptional financing	...	...	...			...	...	...	...	...	...		...	...	
PV of external debt 4/	...	...	17.1			15.2	15.3	15.6	16.1	16.7	17.4		20.6	25.9	
In percent of exports	...	...	89.4			84.4	84.0	84.3	85.9	88.1	90.7		111.4	145.7	
PV of PPG external debt	**...**	**...**	**17.1**			**15.2**	**15.3**	**15.6**	**16.1**	**16.7**	**17.4**		**20.6**	**25.9**	
In percent of exports	**...**	**...**	**89.4**			**84.4**	**84.0**	**84.3**	**85.9**	**88.1**	**90.7**		**111.4**	**145.7**	
In percent of government revenues	**...**	**...**	**95.6**			**86.1**	**89.5**	**89.2**	**90.3**	**91.3**	**92.7**		**103.3**	**124.3**	
Debt service-to-exports ratio (in percent)	**2.1**	**3.0**	**3.0**			**8.3**	**4.9**	**4.5**	**3.9**	**3.5**	**3.3**		**5.1**	**9.3**	
PPG debt service-to-exports ratio (in percent)	**2.1**	**3.0**	**3.0**			**8.3**	**4.9**	**4.5**	**3.9**	**3.5**	**3.3**		**5.1**	**9.3**	
PPG debt service-to-revenue ratio (in percent)	**2.0**	**3.7**	**3.2**			**8.5**	**5.3**	**4.7**	**4.1**	**3.7**	**3.4**		**4.7**	**7.9**	
Total gross financing need (Millions of U.S. dollars)	-4.9	45.3	27.6			46.5	46.0	46.9	48.7	49.7	48.4		84.1	165.5	
Non-interest current account deficit that stabilizes debt ratio	-3.9	3.7	1.8			7.8	5.7	5.6	5.4	5.2	4.6		5.6	4.7	
Key macroeconomic assumptions															
Real GDP growth (in percent)	1.0	2.2	2.7	2.1	0.8	3.4	3.8	4.0	4.2	4.5	4.7	4.1	4.6	4.6	4.6
GDP deflator in US dollar terms (change in percent)	-15.2	2.2	3.3	1.5	8.0	12.5	3.4	3.5	3.0	3.0	2.7	4.7	2.6	2.6	2.6
Effective interest rate (percent) 5/	0.6	0.5	0.2	0.5	0.2	1.1	0.7	0.7	0.8	0.9	1.0	0.9	1.4	1.7	1.5
Growth of exports of G&S (US dollar terms, in percent)	-22.5	13.1	13.4	6.7	14.7	10.0	8.5	9.2	9.0	8.6	8.6	9.0	6.8	6.9	6.9
Growth of imports of G&S (US dollar terms, in percent)	-20.3	0.5	13.8	5.6	14.4	16.9	8.0	7.7	7.4	7.1	7.3	9.1	6.7	6.9	6.8
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	51.9	50.6	42.5	36.6	35.1	35.1	42.0	35.1	35.1	35.1
Government revenues (excluding grants, in percent of GDP)	16.5	14.5	17.8			17.7	17.1	17.5	17.9	18.3	18.7		20.0	20.9	20.2
Aid flows (in Millions of US dollars) 7/	88.9	54.6	76.2			123.4	130.3	138.5	149.0	160.0	173.1		230.7	493.1	
of which: Grants	88.9	54.6	76.2			102.2	108.5	116.7	125.1	134.7	144.8		182.9	365.8	
of which: Concessional loans	0.0	0.0	0.0			21.2	21.9	21.8	23.9	25.3	28.3		47.8	127.3	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			14.9	14.7	14.4	14.2	14.2	14.2		12.8	13.0	12.9
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			91.7	91.7	90.9	89.8	89.7	89.4		86.6	83.3	85.6
Memorandum items:															
Nominal GDP (Millions of US dollars)	588.5	614.2	651.6			757.8	813.8	876.2	940.4	1012.5	1088.9		1554.0	3162.7	
Nominal dollar GDP growth	-14.3	4.4	6.1			16.3	7.4	7.7	7.3	7.7	7.5	9.0	7.4	7.4	7.4
PV of PPG external debt (in Millions of US dollars)			116.5			116.2	125.4	137.0	152.2	169.5	189.4		321.2	821.9	
(PVt-PVt-1)/GDPt-1 (in percent)						-0.1	1.2	1.4	1.7	1.8	2.0	1.4	2.1	2.3	2.2
Gross workers' remittances (Millions of US dollars)	167.8	156.0	169.4			197.2	211.4	226.7	241.9	258.2	277.5		371.3	688.9	
PV of PPG external debt (in percent of GDP + remittances)	...	...	13.5			12.1	12.2	12.4	12.8	13.3	13.8		16.7	21.3	
PV of PPG external debt (in percent of exports + remittances)	...	...	37.8			34.6	34.7	35.1	36.2	37.6	38.9		48.6	65.5	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	1.3			3.4	2.0	1.9	1.6	1.5	1.4		2.2	4.2	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

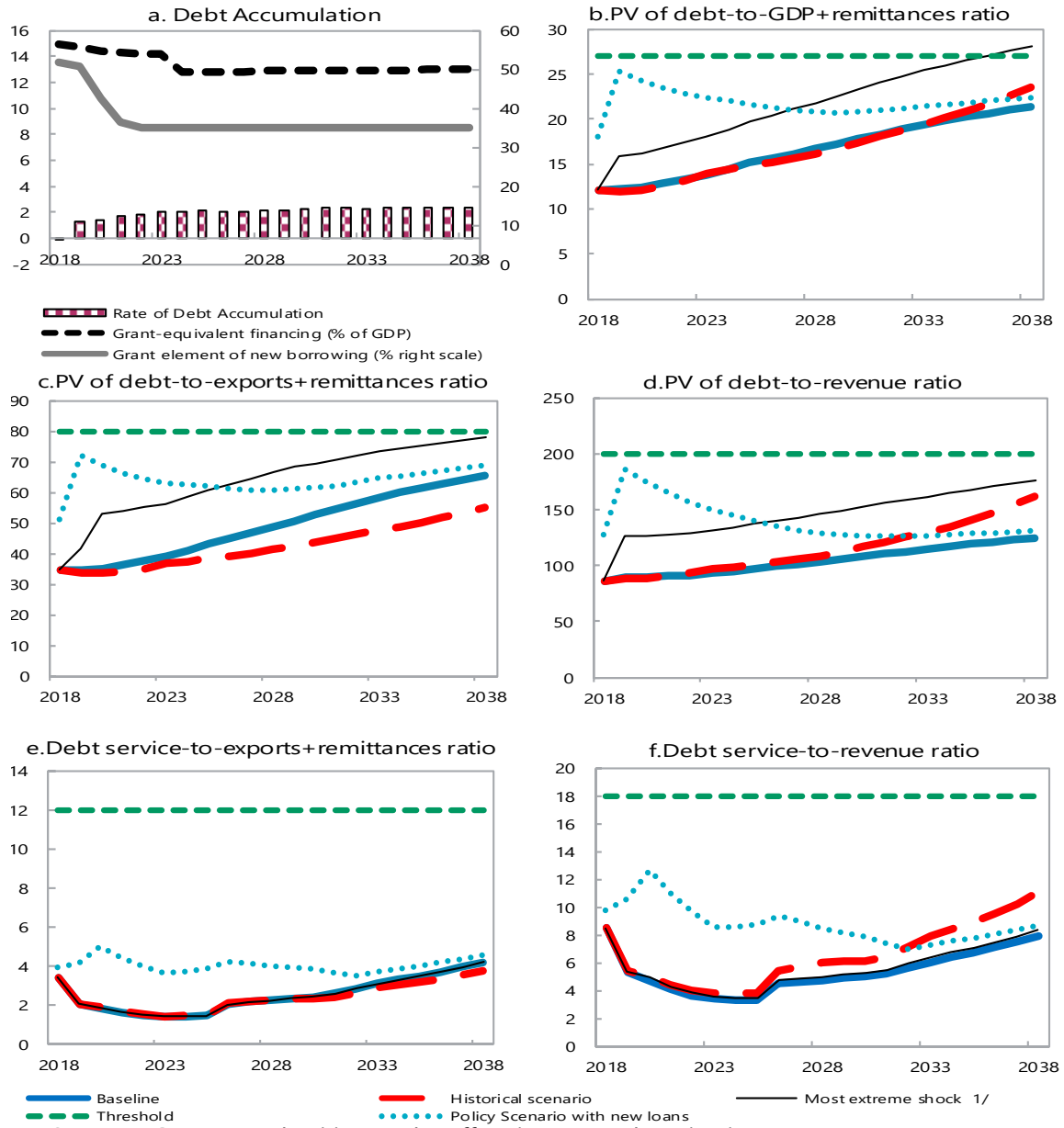
5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Figure 3. Comoros; Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios for the Policy Adjustments Scenarios, 2018-2038 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2028. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Growth shock and in figure f. to a Growth shock

Table 6. Comoros: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt for the Policy Adjustments Scenario, 2018-2038 (in percent)

	Projections							
	2018	2019	2020	2021	2022	2023	**2028**	2038
PV of debt-to-GDP+remittances ratio								
Baseline	12	12	12	13	13	14	**17**	21.3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2018-2038 1/	12	12	12	13	13	14	**16**	24
A2. New public sector loans on less favorable terms in 2018-2038 2	12	13	13	14	15	16	**21**	30
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2019-2020	12	12	13	13	14	14	**17**	22
B2. Export value growth at historical average minus one standard deviation in 2019-2020 3/	12	14	16	17	17	18	**20**	22
B3. US dollar GDP deflator at historical average minus one standard deviation in 2019-2020	12	13	15	15	16	16	**20**	25
B4. Net non-debt creating flows at historical average minus one standard deviation in 2019-2020 4/	12	13	14	14	14	15	**18**	22
B5. Combination of B1-B4 using one-half standard deviation shocks	12	14	18	18	18	18	**21**	24
B6. One-time 30 percent nominal depreciation relative to the baseline in 2019 5/	12	16	16	17	17	18	**22**	28
PV of debt-to-exports+remittances ratio								
Baseline	35	35	35	36	38	39	**49**	66
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2018-2038 1/	35	34	34	34	35	37	**41**	55
A2. New public sector loans on less favorable terms in 2018-2038 2	35	36	38	40	43	45	**61**	93
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2019-2020	35	35	35	36	38	39	**49**	66
B2. Export value growth at historical average minus one standard deviation in 2019-2020 3/	35	41	53	54	55	56	**66**	78
B3. US dollar GDP deflator at historical average minus one standard deviation in 2019-2020	35	35	35	36	38	39	**49**	66
B4. Net non-debt creating flows at historical average minus one standard deviation in 2019-2020 4/	35	37	39	40	41	42	**52**	66
B5. Combination of B1-B4 using one-half standard deviation shocks	35	37	46	44	45	46	**55**	66
B6. One-time 30 percent nominal depreciation relative to the baseline in 2019 5/	35	35	35	36	38	39	**49**	66
PV of debt-to-revenue ratio								
Baseline	86	89	89	90	91	93	**103**	124
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2018-2038 1/	86	89	89	90	93	97	**108**	162
A2. New public sector loans on less favorable terms in 2018-2038 2	86	94	97	101	104	108	**130**	176
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2019-2020	86	92	94	95	96	98	**109**	131
B2. Export value growth at historical average minus one standard deviation in 2019-2020 3/	86	100	119	118	118	118	**124**	129
B3. US dollar GDP deflator at historical average minus one standard deviation in 2019-2020	86	99	109	111	112	114	**126**	152
B4. Net non-debt creating flows at historical average minus one standard deviation in 2019-2020 4/	86	95	98	99	99	100	**109**	126
B5. Combination of B1-B4 using one-half standard deviation shocks	86	104	133	133	133	133	**141**	150
B6. One-time 30 percent nominal depreciation relative to the baseline in 2019 5/	86	127	126	128	129	131	**146**	176

Table 6. Comoros: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt for the Policy Adjustments Scenario, 2018-2038 (in percent) (Concluded)

Debt service-to-exports+remittances ratio								
Baseline	3	2	2	2	2	1	**2**	4.2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2018-2038 1/	3	2	2	2	2	1	**2**	4
A2. New public sector loans on less favorable terms in 2018-2038 2	3	2	2	2	2	2	**3**	6
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2019-2020	3	2	2	2	2	1	**2**	4
B2. Export value growth at historical average minus one standard deviation in 2019-2020 3/	3	2	2	2	2	2	**3**	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2019-2020	3	2	2	2	2	1	**2**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2019-2020 4/	3	2	2	2	2	1	**2**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	3	2	2	2	2	2	**2**	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2019 5/	3	2	2	2	2	1	**2**	4
Debt service-to-revenue ratio								
Baseline	8	5	5	4	4	3	**5**	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2018-2038 1/	8	5	5	4	4	4	**6**	11
A2. New public sector loans on less favorable terms in 2018-2038 2	8	5	5	4	4	4	**6**	12
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2019-2020	8	5	5	4	4	4	**5**	8
B2. Export value growth at historical average minus one standard deviation in 2019-2020 3/	8	5	5	5	4	4	**5**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2019-2020	8	6	6	5	4	4	**6**	10
B4. Net non-debt creating flows at historical average minus one standard deviation in 2019-2020 4/	8	5	5	4	4	4	**5**	8
B5. Combination of B1-B4 using one-half standard deviation shocks	8	6	6	5	5	5	**6**	10
B6. One-time 30 percent nominal depreciation relative to the baseline in 2019 5/	8	7	7	6	5	5	**7**	11
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	36	36	36	36	36	36	**36**	36

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 7. Comoros: Public Sector Debt Sustainability Framework, Policy Adjustments Scenario, 2015-2038 (in percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2015	2016	2017	Average 5/	Standard Deviation 5/	2018	2019	2020	2021	2022	2023	2018-23 Average	2028	2038	2024-38 Average
ublic sector debt 1/	22.8	27.5	29.3			26.9	26.8	26.8	27.0	27.2	27.5		29.8	35.4	
of which: foreign-currency denominated	22.8	26.4	28.8			26.7	26.8	26.8	27.0	27.2	27.5		29.8	35.4	
hange in public sector debt	3.6	4.7	1.8			-2.4	-0.1	0.0	0.2	0.2	0.3		0.5	0.5	
entified debt-creating flows	-2.6	4.2	-4.0			-1.7	0.0	0.5	0.6	0.5	0.4		3.2	4.2	
Primary deficit	-5.9	5.0	-1.2	-3.2	6.5	1.1	1.7	2.2	2.2	2.1	2.1	1.9	4.9	6.1	5.2
Revenue and grants	31.6	23.4	29.5			31.2	30.5	30.8	31.2	31.6	32.0		31.7	32.4	
of which: grants	15.1	8.9	11.7			13.5	13.3	13.3	13.3	13.3	13.3		11.8	11.6	
Primary (noninterest) expenditure	25.7	28.4	28.3			32.2	32.2	33.0	33.4	33.7	34.1		36.6	38.5	
Automatic debt dynamics	3.3	-0.9	-2.8			-2.7	-1.7	-1.7	-1.6	-1.7	-1.6		-1.6	-1.8	
Contribution from interest rate/growth differential	-0.4	-0.9	-1.0			-1.3	-1.4	-1.4	-1.4	-1.5	-1.5		-1.6	-1.8	
of which: contribution from average real interest rate	-0.2	-0.4	-0.3			-0.4	-0.4	-0.4	-0.4	-0.3	-0.3		-0.3	-0.3	
of which: contribution from real GDP growth	-0.2	-0.5	-0.7			-1.0	-1.0	-1.0	-1.1	-1.2	-1.2		-1.3	-1.5	
Contribution from real exchange rate depreciation	3.7	0.0	-1.7			-1.4	-0.3	-0.2	-0.2	-0.1	-0.1		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
esidual, including asset changes	6.2	0.5	5.7			-0.7	-0.1	-0.5	-0.4	-0.3	-0.1		-2.7	-3.7	
her Sustainability Indicators															
/ of public sector debt	...	...	17.6			15.4	15.3	15.6	16.1	16.7	17.4		20.6	25.9	
of which: foreign-currency denominated	...	...	17.1			15.2	15.3	15.6	16.1	16.7	17.4		20.6	25.9	
of which: external	...	...	17.1			15.2	15.3	15.6	16.1	16.7	17.4		20.6	25.9	
/ of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
ross financing need 2/	-5.6	5.6	-0.6			2.6	2.6	3.0	3.0	2.8	2.7		5.8	7.7	
/ of public sector debt-to-revenue and grants ratio (in percent)	...	...	59.5			49.4	50.3	50.6	51.8	52.9	54.2		65.0	80.0	
/ of public sector debt-to-revenue ratio (in percent)	...	...	98.5			87.0	89.5	89.2	90.3	91.3	92.7		103.3	124.3	
of which: external 3/	...	...	95.6			86.1	89.5	89.2	90.3	91.3	92.7		103.3	124.3	
ebt service-to-revenue and grants ratio (in percent) 4/	1.1	2.3	1.9			4.8	3.0	2.7	2.3	2.1	2.0		3.0	5.1	
ebt service-to-revenue ratio (in percent) 4/	2.0	3.7	3.2			8.5	5.3	4.7	4.1	3.7	3.4		4.7	7.9	
imary deficit that stabilizes the debt-to-GDP ratio	-9.5	0.4	-3.0			3.5	1.8	2.2	2.1	1.9	1.7		4.4	5.5	
y macroeconomic and fiscal assumptions															
al GDP growth (in percent)	1.0	2.2	2.7	2.1	0.8	3.4	3.8	4.0	4.2	4.5	4.7	4.1	4.6	4.6	4.6
erage nominal interest rate on forex debt (in percent)	0.6	0.5	0.2	0.5	0.2	1.1	0.7	0.7	0.8	0.9	1.0	0.9	1.4	1.7	1.5
erage real interest rate on domestic debt (in percent)	...	...	...			...	...	...	...	...	...		...	...	
al exchange rate depreciation (in percent, + indicates depreciation)	19.4	0.1	-6.8	1.8	8.1	-5.1	...	...	...	...	...	...	...	...	...
flation rate (GDP deflator, in percent)	1.6	2.4	1.2	3.2	1.4	2.4	2.2	2.3	2.3	2.3	2.3	2.3	2.6	2.6	2.6
owth of real primary spending (deflated by GDP deflator, in percent	7.0	12.8	2.4	2.3	4.3	17.7	3.6	6.6	5.5	5.5	5.9	7.5	5.2	5.2	5.5
ant element of new external borrowing (in percent)	...	...	...	...	...	51.9	50.6	42.5	36.6	35.1	35.1	42.0	35.1	35.1	...

urces: Country authorities; and staff estimates and projections.

[Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

Revenues excluding grants.

Debt service is defined as the sum of interest and amortization of medium and long-term debt.

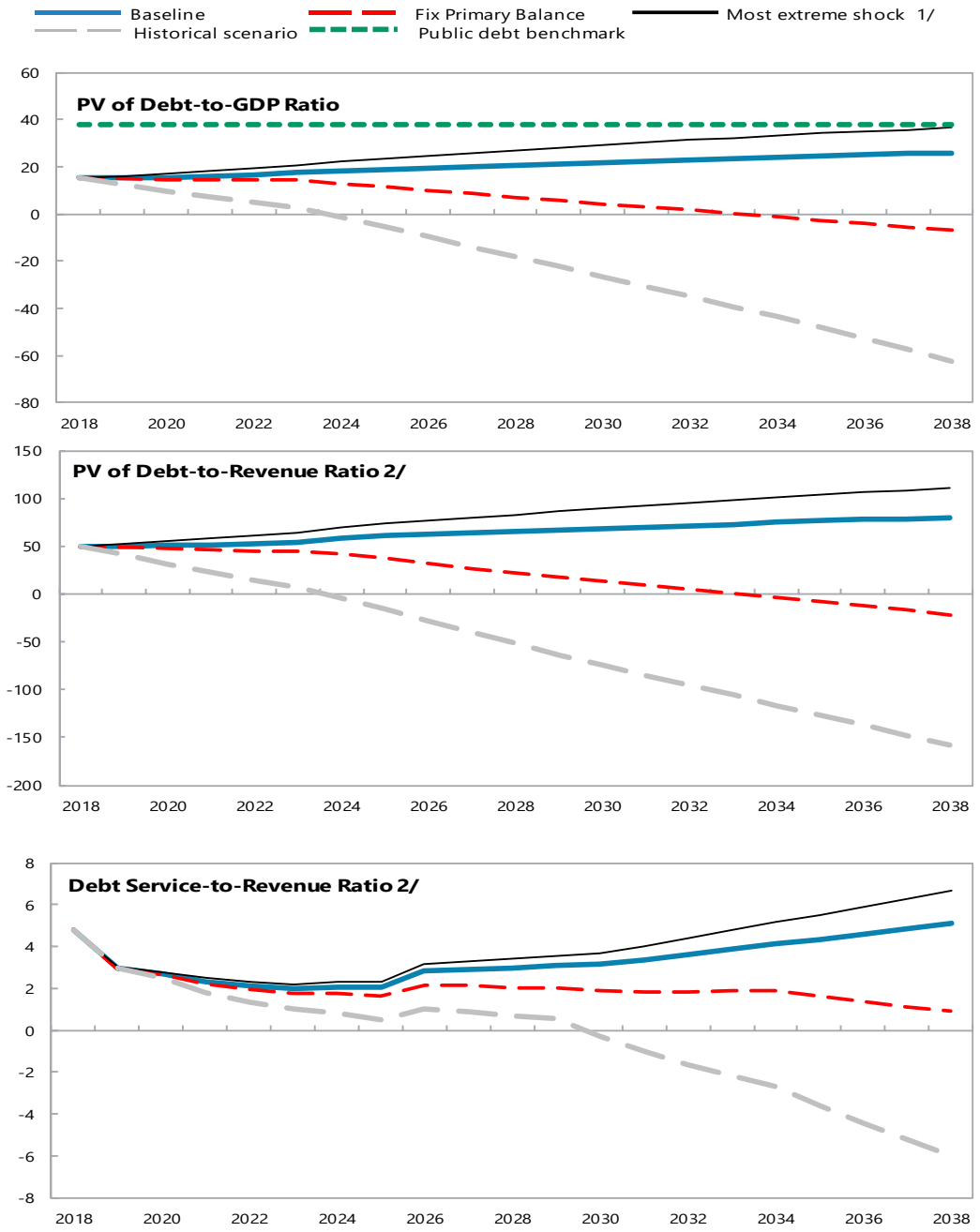
Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 8. Comoros: Sensitivity Analysis for Key Indicators of Public Debt 2018-2038
(policy adjustment scenario)

	Projections							
	2018	2019	2020	2021	2022	2023	2028	2038
PV of Debt-to-GDP Ratio								
Baseline	15	15	16	16	17	17	21	26
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	15	12	10	7	5	3	-18	-62
A2. Primary balance is unchanged from 2018	15	15	14	14	14	14	7	-7
A3. Permanently lower GDP growth 1/	15	15	16	16	17	18	22	31
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2019-20	15	16	17	18	20	21	27	37
B2. Primary balance is at historical average minus one standard deviations in 2019-202	15	16	17	18	18	19	22	27
B3. Combination of B1-B2 using one half standard deviation shocks	15	15	14	15	16	17	23	33
B4. One-time 30 percent real depreciation in 2019	15	21	21	21	21	21	23	25
B5. 10 percent of GDP increase in other debt-creating flows in 2019	15	22	22	22	22	23	25	29
PV of Debt-to-Revenue Ratio 2/								
Baseline	49	50	51	52	53	54	65	80
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	49	41	31	22	14	8	-52	-159
A2. Primary balance is unchanged from 2018	49	49	47	46	45	45	22	-22
A3. Permanently lower GDP growth 1/	49	50	51	52	54	55	69	93
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2019-20	49	52	55	58	60	63	83	111
B2. Primary balance is at historical average minus one standard deviations in 2019-202	49	54	56	57	58	59	69	82
B3. Combination of B1-B2 using one half standard deviation shocks	49	47	44	47	50	53	72	102
B4. One-time 30 percent real depreciation in 2019	49	69	67	66	66	65	71	78
B5. 10 percent of GDP increase in other debt-creating flows in 2019	49	71	71	71	71	72	80	88
Debt Service-to-Revenue Ratio 2/								
Baseline	5	3	3	2	2	2	3	5
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	5	3	2	2	1	1	1	-6
A2. Primary balance is unchanged from 2018	5	3	3	2	2	2	2	1
A3. Permanently lower GDP growth 1/	5	3	3	2	2	2	3	6
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2019-20	5	3	3	2	2	2	3	7
B2. Primary balance is at historical average minus one standard deviations in 2019-202	5	3	3	3	2	2	3	5
B3. Combination of B1-B2 using one half standard deviation shocks	5	3	3	2	2	2	3	6
B4. One-time 30 percent real depreciation in 2019	5	4	4	3	3	3	4	8
B5. 10 percent of GDP increase in other debt-creating flows in 2019	5	3	3	3	3	2	3	6

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Figure 4. Comoros: Indicators of Public Debt Under Alternative Scenarios for the Policy Adjustments Scenario, 2018-2038 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2028.
2/ Revenues are defined inclusive of grants.